

The Mechanism of Surface Passivation Degradation in $\text{SiO}_2/\text{SiN}_x$ Stack Under Light and Elevated Temperature

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Abstract—In this article, a significant degradation of the surface passivation was observed in Czochralski silicon (Cz-Si) lifetime samples using effective lifetime measurements during 1 sun illuminated treatment at 130 °C–175 °C. The samples are passivated by plasma-enhanced chemical vapor deposited silicon nitride (SiN_x) and thermally grown silicon oxide (SiO_2). Samples showed strong variations in the kinetics of the surface passivation degradation depending on the passivation stacks, phosphorus diffusion, and bulk doping. Recovery of the surface passivation was only observed on samples passivated by a single silicon nitride layer. To further investigate the observed behavior, corona charging capacitance-voltage measurements were performed. It is shown that surface degradation arises from a combination of changes in interface defect density (D_{it}) and fixed charge density (Q_f). We show that two separate degradation channels contribute to the degradation in surface passivation, where the fixed charge density continuously declines, while the interface defect density increases to a maximum and subsequently decreases. The decrease in interface defect density leads to a recovery of surface passivation.

Index Terms—Capacitance-voltage (CV), chemical passivation, field-effect passivation, fixed charge, interface defect density, surface passivation degradation.

I. INTRODUCTION

THE past decades have seen dramatic improvements in the efficiency of commercial silicon solar panels. With the increase in cell efficiency, understanding the physics of surface degradation becomes increasingly important to enable the next generation of stable high-efficiency photovoltaic devices and enable cheaper PV energy.

The degradation of the electronic quality of surfaces encompasses hundreds of scientific papers over more than half

a century. More specifically, the degradation of the electronic quality of conventional crystalline silicon solar cells, e.g., passivated emitter and rear cells (PERC), under illumination at room temperature [1], UV illumination [2], storage in the dark [3], or damp-heat conditions [4] is extensive. However, there are fewer data available for their long-term stability under field conditions, i.e., illumination at elevated temperature. Recently, a number of studies have observed that besides light and elevated temperature induced degradation (LeTID), which is a bulk-related degradation, a second degradation phenomenon has been identified as surface passivation degradation and does not appear to be related to UV photons. Compared with other bulk recombination degradation effects, this degradation mechanism requires a longer treatment time to be visible in experimental time scales and is usually observed after full recovery of bulk degradation [5]–[7]. Since then, this degradation mechanism had been observed on a range of passivation layers, including silicon nitride (SiN_x), silicon nitride/silicon oxide ($\text{SiN}_x/\text{SiO}_2$), and silicon nitride/aluminum oxide ($\text{SiN}_x/\text{AlO}_x$) [6], [8]–[12]. The long time scales involved in accelerated surface passivation degradation in the laboratory suggest that surface passivation degradation may take several years before it becomes apparent in the field. However, accelerated laboratory testings are a valid method to predict field degradation [13] and can be applied to predict surface passivation degradation in the field.

In this study, we investigate the surface passivation degradation kinetics in symmetrically passivated lifetime wafers made from Cz-Silicon passivated by thermally grown SiO_2 and plasma-enhanced chemical vapor (PECVD) deposited SiN_x passivation stack (akin to a PERC rear side passivation). A series of illuminated treatments at different high temperatures (130 °C–175 °C) lead to visible degradation of the surface passivation within experimental time scales. Furthermore, the samples were subjected to contactless corona charging capacitance-voltage (CV) measurement after different treatment time scales to examine the underlying physical mechanism for degradation. Analysis of the CV measurements enabled us to separate the relative impact of field-effect passivation and chemical passivation on surface passivation degradation.

II. BACKGROUND

Although researchers have observed surface passivation degradation and its recovery in crystalline silicon wafers with

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a range of passivation layers, both under illumination [5], [7], [9], [12] or annealing in the dark [14], [15], there remain critical knowledge gaps regarding the physics of surface passivation degradation. One of the major areas open for investigation is the cause of the degradation. Sperber *et al.* [5], [8], [9], [12], [16] suggested that surface passivation degradation of SiN_x and SiN_x/SiO₂ passivated wafers occurs as a result of a decrease in the chemical passivation quality. On the other hand, Sperber *et al.* [12] found that surface passivation degradation of SiN_x/AlO_x passivated wafers was because of a decrease in the field-effect quality. However, the value of the interface defect density (D_{it}) and its contribution to the overall surface recombination was not quantified in these works. Although the study from Lin *et al.* [17] demonstrated an increase of interface defect density on AlO_x/SiN_x stack after light soaking, the same study also shows an increase of fixed charge density (Q_f), resulted in an overall increase of the lifetime, consistent with several previous works [18]–[20]. Another work by Liang *et al.* [4] observed surface passivation degradation on SiN_x/AlO_x stack and ascribed it to a loss of chemical passivation, verified by the Fourier transform infrared spectroscopy (FTIR) measurements. The mechanism behind surface recovery also remains unknown. Surface recovery is a difficult issue to investigate in part due to the longer time scales associated with its kinetics.

To study surface passivation degradation, it is necessary to separate bulk from surface-recombination effects. This can be achieved by calculating the saturation current density at the surface (J_{0s}) according to the article presented in [21]. Via J_{0s} measurements, it is possible to calculate a lifetime-equivalent defect density at the surface $\Delta N_{t,s}$, which is typically used to describe surface passivation degradation kinetic, according to the articles presented in [22] and [23]

$$\begin{aligned}\Delta N_{t,s} &= \frac{1}{\tau_{eff}(t)} - \frac{1}{\tau_{eff}(t_0)} \\ &= \frac{2(N_D + \Delta n)}{qWn_i^2} \cdot (J_{0s}(t) - J_{0s}(t_0)) \\ &\propto \Delta J_{0s}.\end{aligned}\quad (1)$$

With N_D being the doping density, Δn is the excess carrier density, W is the wafer thickness, q is the elementary charge, and n_i is the intrinsic charge carrier density. The t_0 marks the sample directly after firing and before any thermal treatment. Whereas $J_{0s}(t)$ marks the J_{0s} value after treatment for the time t , $J_{0s}(t_0)$ marks the J_{0s} value directly after firing.

However, note this terminology is only valid under the assumption that the fixed charge density remains high or has a minor effect on the change of the overall J_{0s} . As reported in [12], the drop in fixed charge density has already been observed. Therefore, the term ΔJ_{0s} is used in this study instead of $\Delta N_{t,s}$, only to ease comparing between samples with different initial surface qualities.

III. EXPERIMENTAL DETAILS

The symmetrical lifetime test samples used in this study were fabricated on commercial grade n-type and p-type Cz-Si wafers with bulk resistivities of $(2.5 \pm 0.2) \Omega \cdot \text{cm}$ and $(1.9 \pm$

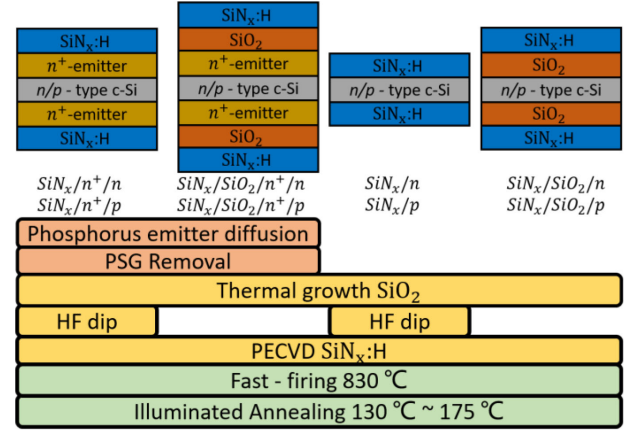


Fig. 1. Fabrication process flow of the investigated samples.

0.1) $\Omega \cdot \text{cm}$, respectively. All as-cut wafers were first treated in an aqueous solution of potassium hydroxide (KOH) ($\sim 22\%$) at 80 °C for saw damage removal, resulting in a final thickness of $(180 \pm 10) \mu\text{m}$. Subsequently, all wafers were cleaned using a sequence developed by Radio Corporation of America (RCA) in preparation for the diffusion or thermal oxide growth process.

As depicted in the process chart (see Fig. 1), selected wafers received a phosphorus emitter diffusion in a phosphoryl chloride (POCl₃) tube diffusion furnace. Phosphorus diffusion was carried out at a temperature of 850 °C for 25 min. The phosphorus glass formed during the diffusion process was subsequently removed by a short dipping in diluted 2% HF solution. All wafers were subsequently thermally oxidized in a tube furnace in an oxygen atmosphere to grow a thin thermal oxide layer. The thermal oxidation process was carried out at 890 °C for 15 min and resulted in a SiO₂ layer thickness of approximately ~ 10 nm. Final resultant sheet resistances of $\sim (46 \pm 1.5 \Omega)/\square$ were measured for the dual-sided phosphorus emitters. Half of the wafers were then dipped in diluted 2% HF solution to remove the oxide layer. All wafers were then RCA clean again prior to dielectric film deposition. A 75 nm SiN_x layer with a refractive index of ~ 2 at 633 nm was then deposited using a remote PECVD reactor (MAiA, Roth and Rau) at a temperature of 400 °C. In the last step, samples were fired at a measured peak temperature of 830 °C using a Schmid infrared inline metallization belt furnace at a speed of 5 m/min. The actual temperature profile of the firing step was measured on an identical sample using a Q18 Datapaq temperature profiler.

The wafers were then laser cleaved into 52×52 mm tokens prior to testing. The testing in this study was conducted in ambient air on hotplates at temperatures ranging from 130 °C to 175 °C under 1 sun equivalent halogen lamp illumination. Injection-dependent effective lifetime τ_{eff} (Δn) measurements were performed after different treatment times using a quasi-steady-state photoconductance (QSSPC) tool (Sinton Instruments, WCT-120TS) at 30 °C. The J_{0s} was extracted at injection level $\Delta n = 10^{16} \text{ cm}^{-3}$, using the Kane–Swanson method [21]. The undiffused samples were additionally analyzed by contactless CV measurements using a commercially available PV-2000 tool (Semilab Company Ltd., Hungary) after different treatment times. This tool uses corona

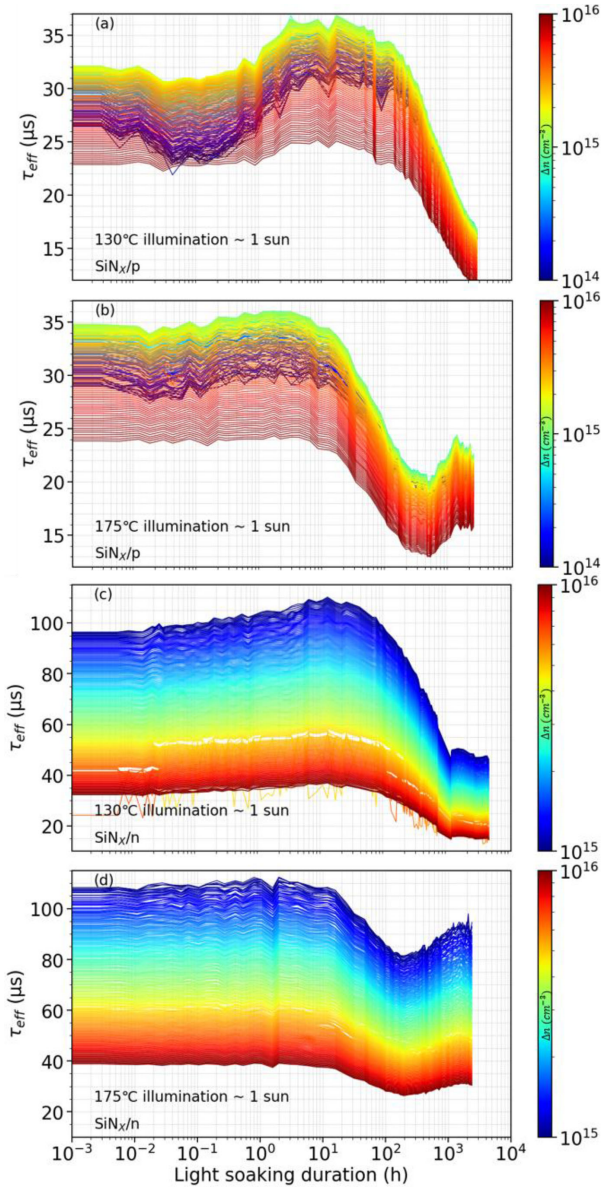


Fig. 2. Time- and injection-resolved τ_{eff} of SiN_x/p and SiN_x/n samples during thermal treatment with ~ 1 sun equivalent illumination at (a) and (c) 130 °C and (b) and (d) 175 °C.

charging to perform contactless CV measurements. The repeated CV measurements were made at the same sample spot on the same sample to minimum uncertainty. After the CV measurement, the samples were dipped in deionized water to remove any remaining charge on the surface and continue testing [12].

IV. RESULTS

A. Overview of the Effective Lifetime Variation Under Illumination and Temperature Treatment

Fig. 2 shows the evolution of the injection-dependent effective lifetime of SiN_x/p and SiN_x/n samples treated under 1 sun illumination with elevated temperatures of 130 °C and 175 °C. At low injection levels, p-type wafers [see Fig. 2(a) and (b)] show a decline of the τ_{eff} followed by a recovery, possibly suggesting a bulk degradation effect. As can be seen, the magnitude of

the degradation decreases as the temperature increases. A possible explanation for this decrease in degradation extent with increasing temperature is mentioned in [15]: bulk degradation rates and recovery rates evolve differently at different temperatures. At higher temperatures, the recovery rate increases faster than the degradation rate, leading to a faster recovery process, therefore, resulting in a less pronounced maximum degradation. The recovered lifetime exceeds the initial lifetime in both cases. In comparison, Fig. 2(c) and (d) shows that the n-type samples are not affected by bulk degradation, which is consistent with previous results [24], [25].

After prolonged illumination treatment, both types of samples exhibit an extensive decline in the effective lifetime over the whole measured injection range. The decline of an effective lifetime at high injection levels indicates a degradation of the surface passivation. The J_0 analysis in Section IV-B and C further confirms the existence of surface passivation degradation. Raising the treatment temperature accelerates the kinetics of the effective lifetime degradation in both n-type and p-type samples. At higher temperatures (175 °C), after reaching maximum degradation, the lifetime begins to increase possibly because of the recovery of the surface passivation quality [5], [9], [12], [14]. The J_0 analysis in Section IV-B and C also confirms this observation with a drop of J_0 .

B. Influence of the Base Material and Surface Doping

To investigate the influence of heavily phosphorus-doped surface and substrate polarity, Fig. 3 depicts a comparison of the evolution in $\Delta J_{0,s}$ on both p-type and n-type samples passivated by SiN_x [see Fig. 3(a)] and $\text{SiN}_x/\text{SiO}_2$ [see Fig. 3(b)]. It can be seen that all samples exhibit a significant rise of $\Delta J_{0,s}$ after approximately 10 h of treatment, confirming that the observed degradation is surface recombination dominated and not bulk recombination dominated. Fig. 3 also indicates both p-type and n-type diffused samples, whether passivated by SiN_x or $\text{SiN}_x/\text{SiO}_2$, all feature a significantly lower extent of degradation than undiffused samples. This arises from the simple fact that the minority carrier concentration at the surface of the diffused samples is orders of magnitude smaller than at the surface of the undiffused samples [7], [12]. Therefore, the samples with highly surface doping are typically less sensitive to change in the interface layer, either reduce of fixed charge density or increase of interface defect density. Note the surface degradation on diffused samples in this work is more significant than that observed in [7]. This is a result of our somewhat lower passivation quality.

Fig. 3 also shows that n-type samples have a higher degradation rate than the p-type samples. A similar phenomenon was demonstrated on accumulation-mode n^+nn^+ MOSFETs, which suffers from more severe interface degradation than inversion-mode n^+pn^+ MOSFETs as a result of interface defect generation [26], [27]. A similar faster reaction on n-type silicon also had been reported before regarding LeTID [15], it was suggested that the lower solubility of hydrogen in n-type silicon than in p-type silicon promotes the generation of the hydrogen-related defect in n-type silicon. As a number of studies have suggested, the cause of surface passivation degradation might also be linked to

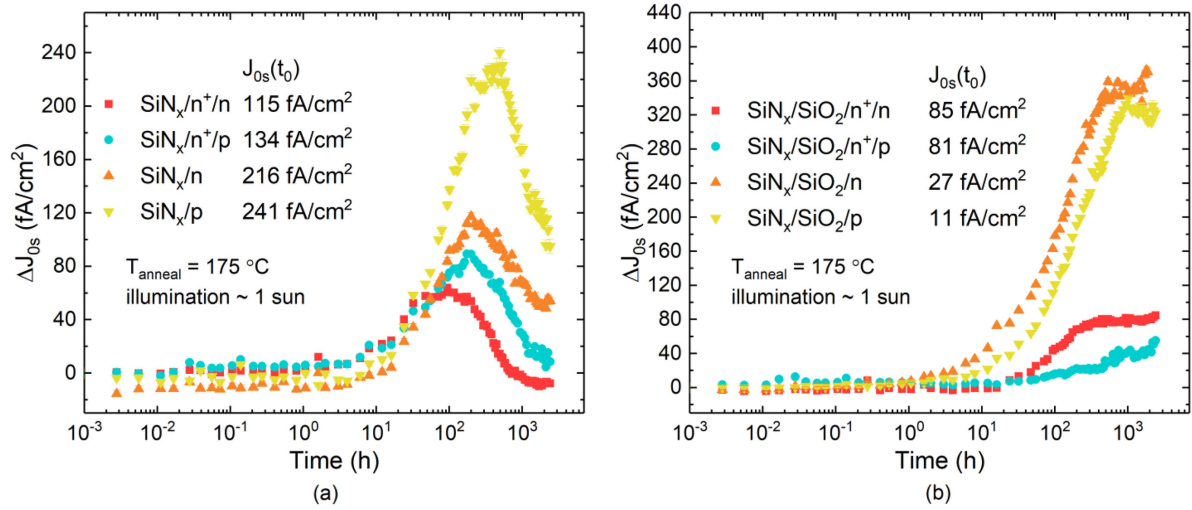


Fig. 3. Time-resolved ΔJ_{0s} ($\Delta n = 10^{16} \text{ cm}^{-3}$) of (a) SiN_x/n, SiN_x/p, SiN_x/n⁺/n, and SiN_x/n⁺/p samples and (b) SiN_x/SiO₂/n, SiN_x/SiO₂/p, SiN_x/SiO₂/n⁺/n, and SiN_x/SiO₂/n⁺/p samples under treatment at 175 °C with ~1 sun equivalent illumination.

hydrogen [5], [7], [14], [15]. However, Chen *et al.* [15] reported a degradation rate constant two orders of magnitude higher in n-type silicon compared with p-type silicon. Note that Sperber *et al.* [9] showed an opposite trend to our work, with p-type FZ-Si wafers having a faster surface degradation than n-type wafers.

C. Comparison of Surface Degradation Between Passivation Layers: SiN_x/SiO₂ Versus SiN_x

Fig. 3 also compares the evolution of the ΔJ_{0s} between SiN_x/SiO₂ and SiN_x passivated wafers. Note that small negative values of ΔJ_{0s} are observed, which point toward a small and fast initial recovery. The first interesting observation is that SiN_x/SiO₂ passivated samples (SiN_x/SiO₂/n⁺/p: 74%, SiN_x/SiO₂/n⁺/n: 94%) show a greater extent of ΔJ_{0s} increase than SiN_x passivated samples (SiN_x/n⁺/p: 67%, SiN_x/n⁺/n: 52%), particularly the undiffused samples (SiN_x/SiO₂/p: 3100%, SiN_x/SiO₂/n: 1300%, SiN_x/p: 100%, SiN_x/n: 56%). A previous study [9] also observed a higher degree of J_{0s} increase on SiN_x/SiO₂ passivated samples (15 times) than SiN_x passivated samples (3 times) under 80 °C and 1 sun illumination treatment. Another interesting observation is that only SiN_x passivated samples show a previously observed recovery of the surface passivation quality [5], [9], [12], [14], while the J_{0s} of SiN_x/SiO₂ passivated samples show no recovery within the experimental time frame.

To examine these observations in more detail, contactless CV measurements were performed on the undiffused samples after various lengths of treatment time to examine the physics behind the surface degradation and recovery mechanisms. Fig. 4 presents the evolution of fixed charge density (Q_f), which reflects the field-effect passivation quality and the evolution of interface defect density (D_{it}) at the midgap as an indicator of chemical passivation quality. Note the $Q_f = Q_{\text{tot}} - Q_{it}$ where the total charge density (Q_{tot}) comes from CV measurement, and the interface charge density (Q_{it}) is calculated as [28]

$$Q_{it} = q \int_{E_v}^{E_c} \{D_{it,D}(E) - D_{it,A}(E)\} f_n(E) dE \quad (2)$$

where $D_{it,D}$ represents the donor-type defect, $D_{it,A}$ represents acceptor-type defect, q is the electron charge, E_c and E_v are the conduction and valence band energy, respectively, and $f_n(E)$ is the electron occupation probability at energy E .

As can be seen in Fig. 4, all samples reveal a different extent of increase in D_{it_midgap} , suggesting various extents in the degradation of the chemical passivation quality (SiN_x/p: 57%, SiN_x/SiO₂/p: 106%, and SiN_x/SiO₂/n: 97%). Note that due to current leakage, the D_{it} measurement is not accurate for the SiN_x/n sample and, therefore, is excluded. After further treatment, the D_{it_midgap} of all samples begins to decline. Fig. 4 also shows a decrease in the field-effect passivation quality with a decrease in the Q_f . However, unlike the trend of D_{it_midgap} , the Q_f shows no recovery but only continuous decline with a maximum relative drop of 66% on SiN_x/SiO₂/p samples and 73% on SiN_x/SiO₂/n samples. Note that SiN_x/p samples also show no Q_f recovery and only a minor maximum relative decline of Q_f (~8.7%) with large fluctuations ($\sim \pm 5\%$) was observed. A similar minor relative decline of Q_f (~16%) was observed previously on SiN_x/p samples [12].

SiN_x/SiO₂ and SiN_x passivated samples share a similar evolution of fixed charge density and interface defect density, as most SiN_x passivated samples typically feature a thin SiO₂ layer (<1 nm) below SiN_x due to oxidation of the surface after wet chemical treatment. Multiple measurements suggest a correlation among change in charge, interface defect density, and variations in J_{0s} for the SiN_x samples. The relative changes in both D_{it_midgap} and Q_f are much larger for SiN_x/SiO₂ passivated samples than for SiN_x passivated samples. This explains the significantly more pronounced degradation of the surface passivation for our samples passivated by SiN_x/SiO₂ (see Fig. 4, black squares).

In both types of samples, the fixed charge density never increases, and thus, we conclude that the J_{0s} recovery observed on SiN_x passivated samples arises as a result of the reduction of the interface defect density. The relative reduction in Q_f is significantly larger for the SiN_x/SiO₂ samples (66%, 73%) than for the SiN_x (8.7%) by an approximate factor of 8. This

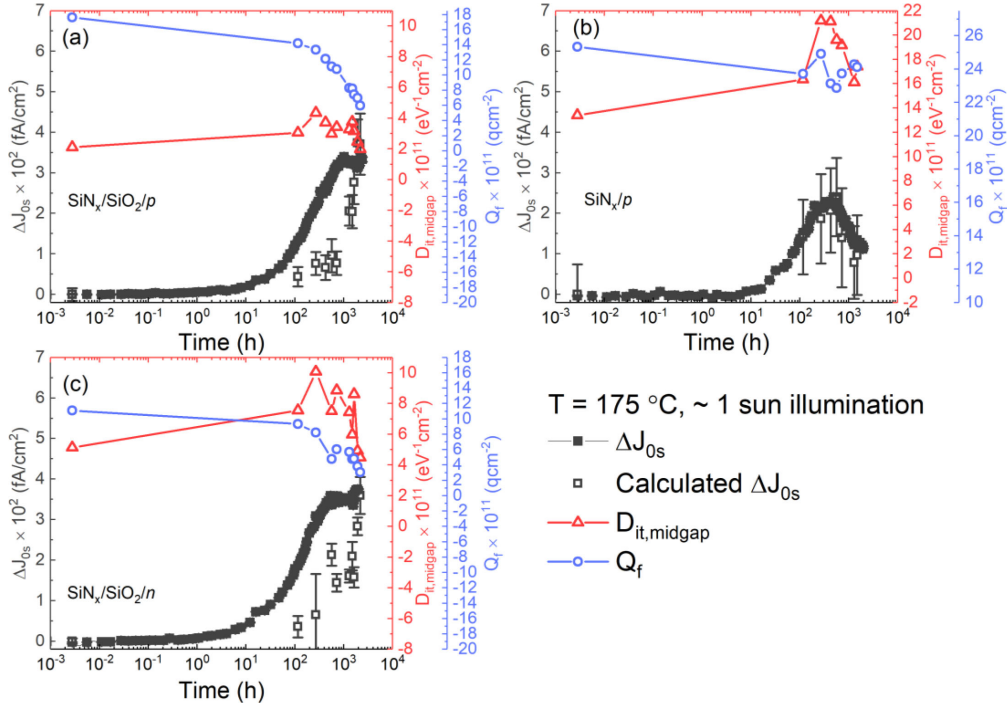


Fig. 4. Series of filled black square marks the time-resolved ΔJ_{0s} ($\Delta n = 10^{16} \text{ cm}^{-3}$) of (a) $\text{SiN}_x/\text{SiO}_2/\text{p}$, (b) SiN_x/p , and (c) $\text{SiN}_x/\text{SiO}_2/\text{n}$ samples during treatment at 175°C and with ~ 1 sun equivalent illumination, measured by the QSSPC tool. The result of CV measurements after different treatment durations is also presented. The series of red triangle marks $D_{it,\text{midgap}}$, while the blue circle marks Q_f . The series of empty black square marks the calculated ΔJ_{0s} from the CV measurements using the calculation method illustrated in the Appendix. Lines serve as guides to the eye.

may compensate for the decrease in D_{it} , explaining the lack of significant recovery in $\text{SiN}_x/\text{SiO}_2$ passivated samples. To examine this possibility, we calculated J_{0s} based on the energy dependence $D_{it}[D_{it}(E)]$ and Q_f using the method proposed by McIntosh and Black [29]. To account for the energy dependence term $D_{it}(E)$, we adopted the parameterization proposed by Xin *et al.* [28], while the energy dependence capture cross section $[\sigma_n(E) \text{ and } \sigma_p(E)]$ was from the articles presented in [30]–[37]. The detailed calculation method is illustrated in the Appendix. The calculated J_{0s} is also presented in Fig. 4 with the opening black symbol. As can be seen, the calculated ΔJ_{0s} displays a similar trend as the measured ΔJ_{0s} , albeit with lower values. The discrepancy between the calculated and measured ΔJ_{0s} arises partly from the assumed capture cross sections and partly from the fitting of the measured $D_{it}(E)$. An example of this fitting can be seen in Fig. 7 in the Appendix. This mismatching may also be due to the small Q_f in our samples and that the measured Q must be smaller or equal to the actual Q_f . As a result, any small absolute errors in Q_f can lead to significant changes in the modeled J_{0s} . In this case, Q_{it} becomes significant and can lead to further uncertainty due to local surface potential/charge fluctuations. Regardless of its absolute value, the calculated ΔJ_{0s} for the $\text{SiN}_x/\text{SiO}_2$ passivated samples shows no decline in agreement with the experimental ΔJ_{0s} measurements.

In summary, we have demonstrated that two separate degradations pathways, namely chemical passivation and field-effect passivation, coexist during the degradation of surface passivation. The different degradation extents of these two mechanisms result in the observed difference J_{0s} evolution behavior between SiN_x passivated samples and $\text{SiN}_x/\text{SiO}_2$ passivated samples.

Our results may indicate that the difference in degradation extent arises from the different thicknesses of the oxide layer, as SiN_x passivated samples usually also feature a thin SiO_2 layer. Further studies are needed to establish the connection between the thickness of the oxide layer and the kinetics and extent of surface degradation.

D. Inadequacy of the Apparent Activation Energy as a General Parameter

Section IV-C demonstrates two separate behavior with the dielectric layer parameters, Q_f and D_{it} , changing simultaneously. It is a common place in the literature to determine the apparent activation energy (E_a) of the surface degradation process [5], [25]. As explained in Section IV-C, the physics of degradation is complex and, thus, this value is not related to a single process but rather the result of a combination in both Q_f and D_{it} at the sample surface. In order to illustrate that a single activation energy is not a suitable parameter to describe the complexity of surface degradation, we calculate the surface degradation and recovery rate as a function of temperature and show their nonlinearity.

In this study, surface degradation and recovery rates are calculated using the following equation:

$$\Delta J_{0s}(t) = \Delta J_{0\text{final}} + \Delta J_{0s,\text{max}} \{ [1 - \exp(-k_{\text{deg}}t)] + [\exp(-k_{\text{rec}}t) - 1] \} \quad (3)$$

where $\Delta J_{0s,\text{max}}$ is the maximum surface degradation, $\Delta J_{0\text{final}}$ is the difference between initial J_0 and final J_0 , and k_{deg} and k_{rec} are the rates of surface degradation and recovery. The model assumes the extracted rates are constant across the experiment,

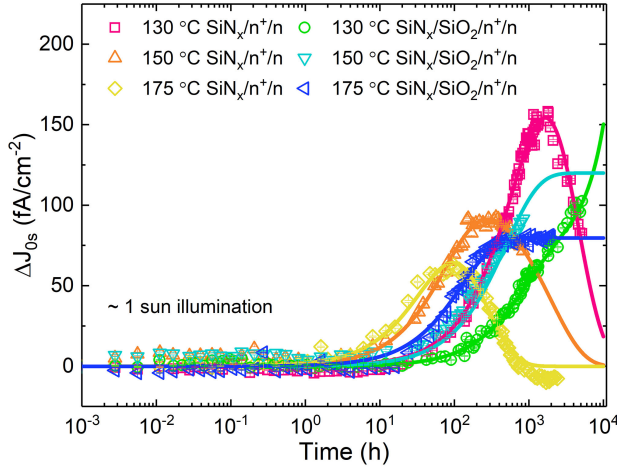


Fig. 5. Time-resolved ΔJ_{0s} (extracted at $\Delta n = 10^{16} \text{ cm}^{-3}$) for an $\text{SiN}_x/\text{n}^+/\text{n}$ and an $\text{SiN}_x/\text{SiO}_2/\text{n}^+/\text{n}$ sample during treatment with ~ 1 sun equivalent illumination at a temperature range between 130°C and 175°C . Solid lines represent modeled fits as given by (3).

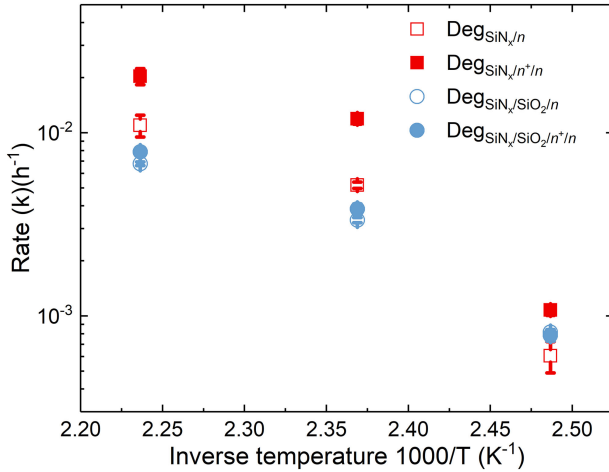


Fig. 6. Degradation rates for n-type silicon under treatment with 1 sun equivalent illumination as a function of the temperature. Square marks samples passivated by a single SiN_x layer and circle marks samples passivated by the $\text{SiN}_x/\text{SiO}_2$ layer. Open symbols mark samples without heavy surface doping and closed symbols mark samples with heavy surface doping.

and all reactions occur simultaneously. The fitted experimental results are shown in Fig. 5. Note, for $\text{SiN}_x/\text{SiO}_2$ passivated samples, or SiN_x passivated samples treated at low temperature without sufficient duration, $k_{\text{rec}} = 0$, as no surface recovery was observed.

Based on the fits, the activation energy for the surface degradation and recovery would usually be extracted using the Arrhenius function defined by

$$k(T) = k_0 \exp\left(-\frac{E_a}{k_B T}\right) \quad (4)$$

where k_B is the Boltzmann constant, T is the treatment temperature in Kelvin, and E_a is the associated activation energy.

The Arrhenius plot of the degradation rate for the n-type samples is shown in Fig. 6. Note, n-type silicon and p-type silicon show similar trends. Although each curve contains only

three points, Fig. 6 indicates that the degradation rate constant does not follow the Arrhenius law under the assumption of a single activation energy. This result makes sense in light of the complex physics involved here with the combination of a change in the field and chemical passivation. More generally, this result indicates that a single activation energy is not an appropriate general parameter to describe surface degradation and recovery. Further work is required to separately calculate the activation energy of the degradation of the surface passivation because of the decline of Q_f and the formation of D_{it} . Additionally, more work is required to investigate the impact of surface orientation on the physics of degradation.

V. CONCLUSION

In this article, we have demonstrated an extended understanding of the mechanisms behind the surface degradation of passivation layers commonly used in silicon solar cells. In Section IV, we demonstrate extensive surface degradation on both p- and n-type substrates, using injection-resolved effective lifetime and J_0 measurements. By comparing SiN_x and $\text{SiN}_x/\text{SiO}_2$ passivated samples, we show degradation in both passivation layers and only observe a surface recovery in SiN_x passivated samples. To unravel the physics of degradation and explain this difference, we perform CV measurements. Our results indicate that surface degradation arises from a combination of a change in interface defect density and fixed charge density where the fixed charge density continuously declines, while the interface defect density increases to a maximum and subsequently decreases. We show that surface recovery is solely because of a decline of interface defect density (chemical passivation). However, for $\text{SiN}_x/\text{SiO}_2$ passivated samples, the improvement in D_{it} (chemical passivation) is not compensated by the decrease in Q_f ; as a result, an effective recovery of the surface passivation is only observed for SiN_x passivated samples. More generally, our results indicate that a single activation energy value for surface degradation kinetics cannot encapsulate the complex physics involved in the degradation of such surfaces. Rather future work may focus on a separate investigation into the D_{it} kinetics (chemical passivation) and Q_f kinetics (potential passivation).

APPENDIX

The calculation, as shown in Fig. 4, was carried out using the method illustrated in [29]. The current density that flows into the surface to recombine is defined by

$$J_{0s} = \frac{qn_i^2}{\frac{p_s + p_1}{S_{n0}} + \frac{n_s + n_1}{S_{p0}}} \quad (A1)$$

where n_i marks the intrinsic carrier concentration, n_s and p_s represent the carrier concentration near the semiconductor surface (interface), the terms p_1 and n_1 incorporate the thermal emission of carriers from the defect energy. S_{p0} and S_{n0} are the surface-recombination velocity parameters for the hole and electron.

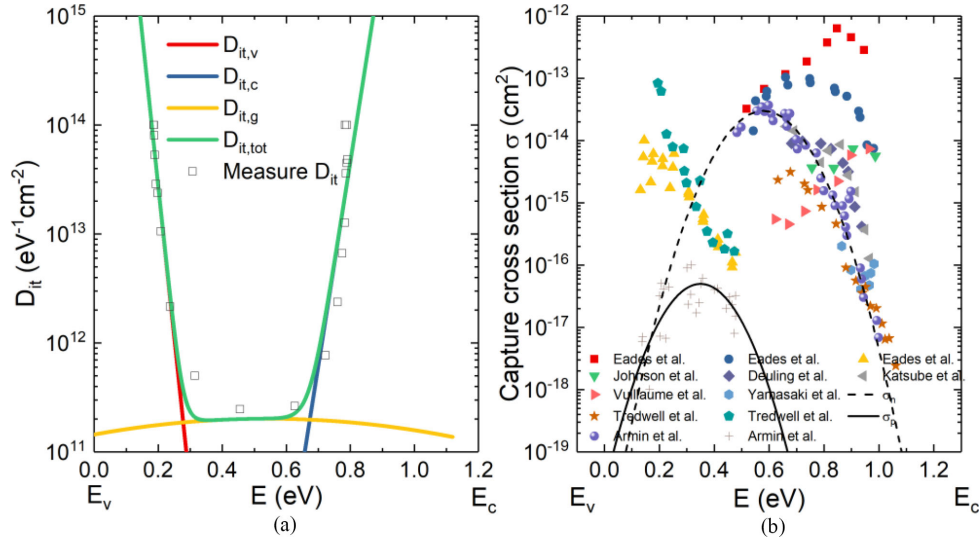


Fig. 7. (a) Example of the measured D_{it} versus energy level and a schematic plot of the three interface defect distributions $D_{it,v}$, $D_{it,c}$, $D_{it,g}$ and their sum $D_{it,tot}$. (b) Survey of the published value of $\sigma_n(E)$ and $\sigma_p(E)$ at the Si-SiO₂ interface [30]–[37]. The lines represent the fit of the data using a normal distribution.

The terms n_1 and p_1 can be calculated as

$$n_1 = N_c \exp\left(-\frac{E_c - E_t}{kT}\right), \quad p_1 = N_v \exp\left(-\frac{E_t - E_v}{kT}\right) \quad (A2)$$

where E_t represents the energy level of the defect state. N_c and N_v represent the effective density of states in the conduction band and the valence band.

The terms S_{p0} and S_{n0} can be calculated as

$$S_{p0} = D_{it} \sigma_p v_{th,p}, \quad S_{n0} = D_{it} \sigma_n v_{th,n}. \quad (A3)$$

The carrier concentrations n_s and p_s can be related to the band bending ψ_s , written as

$$n_s = n_d \exp\left(-\frac{\psi_s}{kT}\right), \quad p_s = p_d \exp\left(\frac{\psi_s}{kT}\right) \quad (A4)$$

where n_d and p_d represent the carrier concentration at the point where the energy bands are unaffected by the charge in the insulator or at the interface. ψ_s is the surface potential relative to $\psi_d = 0$. The well-known charge-neutrality (A5) governs the relationship among the carrier concentrations, band bending, and the surface charge

$$p_s + n_s - p_d - p_s + (N_A + N_D) \frac{\psi_s}{kT} = \frac{Q^2}{2q\epsilon_{si}kT}. \quad (A5)$$

Thus, as can be seen above, by numerically solving the implicit function (A5), $J_{0,s}$ can be calculated from two important dielectric layer parameters: Q_f and D_{it} .

The defect states at a semiconductor surface are typically a distribution of defect energies across the bandgap. To account for energy dependences' terms, such as $n_1(E)$, $p_1(E)$, and $D_{it}(E)$, an integrated approach is often applied. When this is the case, the surface recombination (U_s) can be described as

$$U_s = \int_{E_v}^{E_c} \frac{p_s n_s - n_i^2}{\frac{p_s + p_1(E)}{S_{n0}(E)} + \frac{n_s + n_1(E)}{S_{p0}(E)}} dE \quad (A6)$$

$$S_{p0} = D_{it}(E) \sigma_p(E) V_{th,p}, \quad S_{n0} = D_{it}(E) \sigma_n(E) V_{th,n}. \quad (A7)$$

Among these terms, $D_{it}(E)$ usually exhibits a U-shaped energy dependence, a behavior that is typically reported by Xin *et al.* [28], [34], [38]. Fig. 7(a) shows an example of the measured $D_{it}(E)$ in this work. To account for this U-shaped energy dependence, this study applied a method as proposed by Xin *et al.* [28], fitted the measured $D_{it}(E)$ curves using a near midgap Gaussian defect distribution and two exponential tail state distributions, which are increasing toward the edges of the bandgap, as can be seen in Fig. 7(a).

The fitting (A8)–(A11) define the exact shape of the total interface density distribution $D_{it,tot}(E)$. Note that σ^2 represents the variance of the Gaussian distribution, and E_0 represents the energy located at the peak of the Gaussian distribution for $D_{it,g}$.

$$D_{it,v}(E) = D_{it,0v} \times e^{\frac{E_v - E}{E_v - trap}} \quad (A8)$$

$$D_{it,c}(E) = D_{it,0c} \times e^{\frac{E - E_c}{E_c - trap}} \quad (A9)$$

$$D_{it,g}(E) = D_{it,0g} \times e^{-\frac{1}{2}\left(\frac{E - E_0}{\sigma}\right)^2} \quad (A10)$$

$$D_{it,tot}(E) = D_{it,v}(E) + D_{it,g}(E) + D_{it,c}(E). \quad (A11)$$

The terms $\sigma_n(E)$ and $\sigma_p(E)$ are obtained from the values for the capture cross sections at the Si-SiO₂ interface reported in the literature [30]–[37], scattered over a wide range of energy level, as shown in Fig. 7(b). The terms $\sigma_n(E)$ and $\sigma_p(E)$ are modeled using a Gaussian distribution

$$\sigma_{\frac{n}{p}}(E) = \sigma_0 \times e^{-\frac{1}{2}\left(\frac{E - E_0, \frac{n}{p}}{\mu}\right)^2}. \quad (A12)$$

The fitting parameters are given in Table I.

TABLE I
FITTED PARAMETERS FOR THE PROPOSED $\sigma_n(E)$ AND $\sigma_p(E)$ DISTRIBUTION

	σ_0	μ	E_0
σ_n	3×10^{-14}	0.1	0.58
σ_p	5×10^{-17}	0.09	0.35

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